

TECHNICAL SPECIFICATION



**Photovoltaic (PV) modules – Type approval, design and safety qualification –
Retesting**



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IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
info@iec.ch
www.iec.ch

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TECHNICAL SPECIFICATION



Photovoltaic (PV) modules – Type approval, design and safety qualification – Retesting

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**PHOTOVOLTAIC (PV) MODULES – TYPE APPROVAL,
DESIGN AND SAFETY QUALIFICATION – RETESTING**

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Technical specifications are subject to review within three years of publication to decide whether they can be transformed into International Standards.

IEC TS 62915, which is a technical specification, has been prepared by IEC technical committee 82: Solar photovoltaic energy systems.

The text of this technical specification is based on the following documents:

Enquiry draft	Reports on voting
82/1331/DTS	82/1378A/RVDTS

Full information on the voting for the approval of this technical specification can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- transformed into an International standard,
- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

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PHOTOVOLTAIC (PV) MODULES – TYPE APPROVAL, DESIGN AND SAFETY QUALIFICATION – RETESTING

1 Scope

This document sets forth a uniform approach to maintain type approval, design and safety qualification of terrestrial PV modules that have undergone, or will undergo modification from their originally assessed design.

Changes in material selection, components and manufacturing process can impact electrical performance, reliability and safety of the modified product. This document lists typical modifications and the resulting requirements for retesting based on the different test standards. It provides assistance; at some level engineering judgement may be needed.

The test sequences are selected to identify adverse changes to the modified product.

Those products successfully following the herein defined test sequences are considered to be compliant with the standard against which they have originally been assessed in a full qualification.

The number of samples to be included in the retesting program and the pass/fail criteria are listed in the referenced standards IEC 61215 and IEC 61730.

Tests required by changes from previous to new standard editions of IEC 61215 and IEC 61730 are not covered by this document and are evaluated separately.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61215 (all parts), *Terrestrial photovoltaic (PV) modules – Design qualification and type approval*

IEC 61215-1:2016, *Terrestrial photovoltaic (PV) modules – Design qualification and type approval – Part 1: Test requirements*

IEC 61215-2:2016, *Terrestrial photovoltaic (PV) modules – Design qualification and type approval – Part 2: Test procedures*

IEC 61730 (all parts), *Photovoltaic (PV) module safety qualification*

IEC 61730-1:2016, *Photovoltaic (PV) module safety qualification – Part 1: Requirements for construction*

IEC TS 61836, *Solar photovoltaic energy systems – Terms, definitions and symbols*

IEC 62790, *Junction boxes for photovoltaic modules – Safety requirements and tests*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 61215-1, IEC 61215-2, IEC 61730-1 and IEC TS 61836, as well as the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1

different material

material that differs in its chemical composition, type designation, or specification from the material it replaces

3.2

nominal value

value of a quantity used to designate and identify a component, device, equipment, or system

[SOURCE IEC 60050-151:2001, 151-16-09]

3.3

tolerance

permitted deviation of declared nominal value

4 Retesting

4.1 General

This clause is separated into one subclause each for crystalline silicon and for thin-film technologies. The document is organized by major modification headings with specific supporting examples and parenthetical reference to the specific clauses of the relevant IEC standards.

Any change in the design, materials, components, material combinations, manufacturers or processing of the PV module type family from the last tested version may require a repetition of some or all of the qualification tests according to the clauses that follow in order to maintain type and safety approval. For any change in material specification, including, e.g., electrical, optical, mechanical properties, the nominal values and tolerances shall be considered. For any assessment of a new thickness or dimension, the initially tested thickness or dimension shall be used as reference. Any variation of a parameter may be assessed as change if the new value is out of the tolerance from the nominal value of this parameter.

Materials in direct contact with each other shall be tested in all applicable combinations. The required test items shall be selected only from those tests which are applicable for change of both materials. An example for an assessment procedure is given in Annex A, Clause A.4.

The number of samples to be included in the retesting program and the pass criteria are to be taken from the relevant clause/subclause of the referenced standards ('pass criteria').

Each PV module delivered for retesting shall be subjected to electrical stabilisation (MQT 19), as applicable by the relevant type approval standard.

All initial measurements as listed in the referenced standards shall be performed before the specific tests, e.g. tests MQT 01 / 03 / 06.1 / 15 / 19 for an IEC 61215 retest program.

Any scenario including a change in the optical path or electric circuitry that requires retesting as defined hereinafter shall include an STC output power measurement (MQT 06.1). The measured stabilized power, open-circuit voltage and short-circuit current shall be assessed against the rating (Gate No. 1), and the relative change in output power shall be assessed (Gate No. 2) according to the pass criteria laid down in the standard (see IEC 61215-1:2016, 7.2).

Final diagnostic measurements are listed in the referenced test procedure; as a minimum the same tests as performed initially shall be performed.

The Durability of markings (MST 05) and the Sharp edge test (MST 06) need to be considered in general for all design changes which may impact the results of these tests.

If multiple tests from a test sequence are required, they shall be done in the sequence prescribed by the referenced standard.

Changes in the PV module design might require assessment against IEC 61730-1 (requirements for construction) besides the indicated test programs.

Required tests in this Clause 4 are written for combined IEC 61215 (all parts) and IEC 61730 (all parts) evaluations. For simplification, the term “all parts” is omitted in the following. For single IEC 61730 evaluations, care has to be taken that tests listed herein for IEC 61215 may also be referenced and required by IEC 61730 to ensure compliance.

4.2 Test programs for crystalline silicon PV modules

NOTE See Table A.1 for a summary of the retest requirements for crystalline silicon PV modules.

4.2.1 Modification to frontsheet

A change from glass to non-glass or vice-versa requires a full qualification.

For the following modifications:

- Different material, i.e. any change in specification of the material or any of its layers
- Glass: reduction of thickness by more than 10 %; non-glass: change of thickness by more than 20 % of any one of the individual layers (while maintaining the required minimum distance through insulation)
- For glass, if there is a reduction in the strengthening process (for example retest if change is from tempered glass to heat strengthened or annealed)
- Different surface treatment, e.g. any coating on frontsheet (inside or outside)
- Change of amount of adhesives, primers or other additives
- Addition or removal of adhesives, primers or additives

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09) if change in material, heat strengthening process or if thickness is reduced
- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) / Retention of junction box on mounting surface (MQT 14.1) (can be omitted for glass with identical UV cut-off)
- Damp heat test (MQT 13) if non-glass or if surface treatment is added/changed (inside or outside)
- Static mechanical load test (MQT 16) (can be omitted for different inside and outside surface treatments that do not impair mechanical strength)
- Hail test (MQT 17) (can be omitted for different surface treatment on the inside)

Repeat for IEC 61730:

- Insulation thickness test (MST 04) if non-glass
- Cut susceptibility test (MST 12) if non-glass
- Impulse voltage test (MST 14) if reduced thickness or if change in material
- Temperature test (MST 21) if non-glass and change in material
- Ignitability test (MST 24) if non-glass
- Module breakage test (MST 32) (can be omitted for different surface treatments that do not impair mechanical strength)
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint (not for reduction of thickness, not for different outer surface treatment and not for change in glass strengthening process)
- Materials creep test (MST 37) (not for reduction of thickness and not for different outside surface treatment)
- Sequence B if non-glass
- Sequence B1 if design qualified for pollution degree 1 (not for reduction of thickness, not for different surface treatment and not for change in glass strengthening process)

For increased thickness, the Materials creep test (MST 37) is required.

4.2.2 Modification to encapsulation system

For the following modifications:

- Different material
- Different type or change in amount of additive or different chemical composition of encapsulant
- Different manufacturer of encapsulant
- Different encapsulation process (e.g. curing degree, temperature/pressure profile)
- Reduction in thickness of total encapsulation by more than 20 % prior to processing (thickness can also be expressed in density, e.g. in g/cm³)

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12)
- Damp heat test (MQT 13)
- Hail test (MQT 17) if frontsheet is polymeric

Repeat for IEC 61730:

- Cut susceptibility test (MST 12) if frontsheet or backsheet is polymeric
- Impulse voltage test (MST 14) if reduced thickness or if change in material
- Module breakage test (MST 32) if material composition changes
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes encapsulant as a part of a qualified cemented joint
- Materials creep test (MST 37)
- Sequence B (only for different material or reduction in thickness)
- Sequence B1 if design qualified for pollution degree 1

4.2.3 Modification to cell technology

For the following modifications:

- Metallization material composition (e.g. paste)
- Change in busbar metallization area by more than 20 %
- Change in number of busbars
- Change in anti-reflective coating
- Semiconductor layer material
- Change in crystallization process (e.g. mono- vs. poly-crystalline)
- Change of manufacturing site of the solar cells not under the same quality management system
- Use of cells from a different manufacturer
- Change in nominal cell thickness greater than 10 %
- Different size of cell or use of cut cells (e.g. halved)

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- Thermal cycling test, 200 cycles (MQT 11)
- Damp heat test (MQT 13) (may be omitted if outer surface of cell is chemically identical (metallization and AR coating))
- Static mechanical load test (MQT 16) for reduction of cell thickness only

Repeat for IEC 61730:

- Temperature test (MST 21)
- Reverse current overload test (MST 26)

4.2.4 Modification to cell and string interconnect material or technique

For the following modifications:

- Different material (e.g. alloy, chemistry and core)
- Change in mechanical properties by more than 10 % of tensile strength, yield strength and elongation
- Change in thickness by more than 10 %
- Change in (total) cross-section of interconnect material (e.g., more busbars / more busbars with less width)
- Different bonding technique
- Change in the number of interconnect or bonding points or decrease in bonding area per contact point
- Different length of interconnect material between last bond on one cell and first bond on the adjacent cell
- Different solder material, flux or conductive adhesive
- Change in insulation tape (thickness, material, manufacturer)

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09) for changes in bonding technique, interconnect material, solder material, flux or conductive adhesive
- Thermal cycling test, 200 cycles (MQT 11)

- Damp heat test (MQT 13) for changes in material

Repeat for IEC 61730:

- Reverse current overload test (MST 26)

4.2.5 Modification to backsheet

A change from glass to non-glass or vice-versa requires a full qualification.

For the following modifications:

- Different material, i.e. any change in specification of the material or any of its layers
- Glass: reduction of thickness by more than 10 %; non-glass: change of thickness by more than 20 % of any one of the individual layers (while maintaining the required minimum distance through insulation)
- For glass, if there is a reduction in the strengthening process (e.g. retest if change is from tempered to heat strengthened or annealed glass)
- Different surface treatments (inside or outside)
- Change of amount of adhesives, primers or other additives
- Addition or removal of adhesives, primers or additives

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09) for glass if change in heat strengthening process or if thickness is reduced
- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) / Retention of junction box on mounting surface (MQT 14.1) (can be omitted for glass with identical UV cut-off; not for glass change). MQT 14.1 can be omitted if junction box is mounted on frontsheet.
- Damp heat test (MQT 13) if non-glass or if surface treatment is added/changed (inside or outside).
- Static mechanical load test (MQT 16) if glass (including change in manufacturer) or if mounting depends on adhesion to backsheet
- Hail test (MQT 17) if rigidity depends on backsheet

Repeat for IEC 61730:

- Insulation thickness test (MST 04) if non-glass
- Cut susceptibility test (MST 12) if non-glass
- Impulse voltage test (MST 14) if reduced thickness or if change in material
- Temperature test (MST 21) if non-glass and if change in material
- Ignitability test (MST 24) if non-glass
- Module breakage test (MST 32) if glass (can be omitted for different surface treatments that do not impair mechanical strength)
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint and if backsheet is part of it
- Materials creep test (MST 37) (not for reduction of thickness and not for different outside surface treatment)
- Sequence B if non-glass
- Sequence B1 if design qualified for pollution degree 1

For increased thickness, the Materials creep test (MST 37) is required.

Additionally, in case of colour change of backsheet potentially resulting in higher PV module operating temperatures, consider repetition of the Temperature test (MST 21), if applicable.

4.2.6 Modification to electrical termination

Components for electrical terminations such as junction box, cables and connectors shall meet the relevant IEC standards referenced in IEC 61730-1. Their combination with other components and materials shall be tested either on PV module level or on component level. A change in component combination may result in the following PV module tests in addition to tests listed in the relevant component standards.

For the following modifications:

- Different material
- Different design (e.g., different dimensions, changed positions, number of junction boxes)
- Different potting material
- Different method of mechanical attachment / securement (e.g. adhesive change)
- Different method of electrical attachment (e.g. solder, crimping, brazing, etc.)

Repeat for IEC 61215:

- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) / Robustness of terminations test (MQT 14.1 and 14.2) (UV preconditioning test can be omitted for change in potting material or in case junction box is not directly exposed to sunlight; Test of cord anchorage (MQT 14.2) can be omitted for change in mechanical attachment of junction box; Retention of junction box on mounting surface (MQT 14.1) can be omitted for change in electrical attachment of cables)
- Thermal cycling test, 200 cycles (MQT 11) only for change in electrical attachment
- Damp heat test (MQT 13)
- Bypass diode thermal test (MQT 18) (not required for change of any attachment)

Repeat for IEC 61730:

- Accessibility test (MST 11)
- Temperature test (MST 21) if change in potting material or adhesive
- Ignitability test (MST 24) only for change of adhesive
- Reverse current overload test (MST 26) (not for change of adhesive)
- Screw connections test (MST 33) if applicable
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint (only for mechanical attachment of junction box)
- Materials creep test (MST 37) only for change of adhesive or for increased weight of electrical termination
- Sequence B only for change of adhesive
- Sequence B1 if design qualified for pollution degree 1

4.2.7 Modification to bypass diode

For bypass diodes mounted within the junction box, requirements of IEC 62790 need to be fulfilled.

For bypass diodes not mounted within the junction box, the following applies:

For the following modifications:

- Lower rating of diode current or diode junction temperature
- Different number of bypass diodes per PV module
- Different type of bypass diode
- Different manufacturer of bypass diode
- Different mounting method (physical configuration, soldering material, bonding process, soldering temperature or process)

Repeat for IEC 61215:

- Thermal cycling test, 200 cycles (MQT 11) only for different mounting method
- Bypass diode thermal test (MQT 18)

Repeat for IEC 61730:

- Reverse current overload test (MST 26) only for different mounting method

4.2.8 Modification to electrical circuitry

For the following modifications:

- Modifications to the interconnection circuitry (e.g. more cells per bypass diode or rerouting of output leads)
- Reconfiguration of PV module operating voltage/current (e.g. serial/parallel connection of cells)

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09) only if more cells per bypass diode
- Thermal cycling test, 200 cycles (MQT 11) if there are internal conductors behind the cells
- Bypass diode thermal test (MQT 18) if the short circuit current increases by >10 %

Repeat for IEC 61730:

- Insulation thickness test (MST 04) for rerouting of output leads
- Reverse current overload test (MST 26) (only for increase in PV module operating voltage/current by 10 % or more)

4.2.9 Modification to edge sealing

For the following modifications:

- Different material
- Different thickness or width

Repeat for IEC 61215:

- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) if edge sealing is outer enclosure
- Damp heat test (MQT 13)

Repeat for IEC 61730:

- Impulse voltage test (MST 14)
- Ignitability test (MST 24) (not for different thickness or width)
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint
- Sequence B (not for different thickness or width)

- Sequence B1 if design qualified for pollution degree 1

4.2.10 Modification to frame and/or mounting structure

For the following modifications:

- Shape and/or cross-section of frame
- Reduction of surface area in contact between laminate and frame per linear dimension
- Different material including adhesive or mounting material
- Different mounting method (as defined in installation manual)
- Change in frame corner design
- Change in frame adhesive
- Change from framed to frameless PV module or vice versa

Repeat for IEC 61215:

- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) if mounting relies on adhesive or polymeric framing material
- Thermal cycling test, 200 cycles (MQT 11) if mounting relies on adhesive or polymeric framing material
- Damp heat test (MQT 13) if mounting relies on adhesive or polymeric framing material or if change from framed to frameless PV module or vice versa
- Static mechanical load test (MQT 16)
- Hail test (MQT 17) if polymeric frame / frontsheet or if change from framed to frameless PV module

Repeat for IEC 61730:

- Continuity test of equipotential bonding (MST 13) if change in method of assembly (can be omitted if change in adhesive)
- Ignitability test (MST 24) for polymeric frames
- Module breakage test (MST 32)
- Screw connections test (MST 33) if applicable
- Material creep test (MST 37) if creep is not prevented by frame or other support anymore
- Sequence B for polymeric frames

In case of change of frame manufacturer or mounting system manufacturer (same material specifications and design) no tests are required.

4.2.11 Change in PV module size

For increase by more than 20 % of length, width or area

Repeat for IEC 61215:

- Thermal cycling test, 200 cycles (MQT 11)
- Damp heat test (MQT 13)
- Static mechanical load test (MQT 16)
- Hail test (MQT 17) if non-tempered glass or if non-glass

Repeat for IEC 61730:

- Module breakage test (MST 32)

4.2.12 Higher or lower output power (by 10 % or more) with the identical design and size and using the identical cell process

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- Thermal cycling test, 200 cycles (MQT 11) if short-circuit current is increased by more than 10 %
- Bypass diode thermal test (MQT 18) if short-circuit current is increased by more than 10 %

For performance at STC (MQT 06.1) see 4.1.

Repeat for IEC 61730:

- Reverse current overload test (MST 26)

4.2.13 Increase of over-current protection rating

Repeat for IEC 61730:

- Continuity test of equipotential bonding (MST 13)
- Reverse current overload test (MST 26)

4.2.14 Increase of system voltage

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12)
- Thermal cycling test, 200 cycles (MQT 11)
- Damp heat test (MQT 13)

Repeat for IEC 61730:

- Insulation thickness test (MST 04)
- Accessibility test (MST 11)
- Cut susceptibility test (MST 12) if non-glass
- Continuity test of equipotential bonding (MST 13)
- Impulse voltage test (MST 14)
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint
- Sequence B

4.2.15 Change in cell fixing tape

For the following modifications:

- Different material
- Different manufacturer

Repeat for IEC 61215:

- Humidity freeze test (MQT 12)

4.3 Test programs for thin-film PV modules

NOTE See Table A.2 for a summary of the retest requirements for thin-film PV modules.

4.3.1 Modification to frontsheet

A change from glass to non-glass or vice-versa requires a full qualification.

For the following modifications:

- Different material, i.e. any change in specification of the material or any of its layers
- Glass: reduction of thickness by more than 10 %; non-glass: change of thickness by more than 20 % of any one of the individual layers (while maintaining the required minimum distance through insulation)
- For glass, if there is a reduction in the strengthening process (for example retest if change is from tempered glass to heat strengthened or annealed)
- Different surface treatment, e.g. any coating on frontsheet (inside or outside)
- Change of amount of adhesives, primers or other additives
- Addition or removal of adhesives, primers or additives

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09) if change in material, heat strengthening process or if thickness is reduced
- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) / Retention of junction box on mounting surface (MQT 14.1) (can be omitted for glass with identical UV cut-off)
- Damp heat test (MQT 13) if non-glass or if cell material is deposited on glass or if surface treatment is added/changed (inside or outside)
- Static mechanical load test (MQT 16) (can be omitted for different inside and outside surface treatment that do not impair mechanical strength)
- Hail test (MQT 17) (can be omitted for different surface treatment on the inside)

Repeat for IEC 61730:

- Insulation thickness test (MST 04) if non-glass
- Cut susceptibility test (MST 12) if non-glass
- Impulse voltage test (MST 14) if reduced thickness or if change in material
- Temperature test (MST 21) if non-glass and change in material
- Ignitability test (MST 24) if non-glass
- Module breakage test (MST 32) (can be omitted for different surface treatments that do not impair mechanical strength)
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint (not for reduction of thickness, not for different outer surface treatment and not for change in glass strengthening process)
- Materials creep test (MST 37) (not for reduction of thickness and not for different outside surface treatment)
- Sequence B if non-glass
- Sequence B1 if design qualified for pollution degree 1 (not for reduction of thickness, not for different surface treatment and not for change in glass strengthening process)

For increased thickness, the Materials creep test (MST 37) is required.

4.3.2 Modification to encapsulation system

For the following modifications:

- Different material
- Different type or change in amount of additive or different chemical composition of encapsulant
- Different manufacturer of encapsulant
- Different encapsulation process (e.g. curing degree, temperature/pressure profile)
- Reduction in thickness of total encapsulation by more than 20 % prior to processing (thickness can also be expressed in density, e.g. in g/cm³)

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12)
- Damp heat test (MQT 13)
- Hail test (MQT 17) if frontsheet is polymeric

Repeat for IEC 61730:

- Cut susceptibility test (MST 12) if frontsheet or backsheet is polymeric
- Impulse voltage test (MST 14) if reduced thickness or if change in material
- Module breakage test (MST 32) if material composition changes
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes encapsulant as a part of a qualified cemented joint
- Materials creep test (MST 37)
- Sequence B (only for different material or reduction in thickness)
- Sequence B1 if design qualified for pollution degree 1

4.3.3 Modification to front contact (e. g. TCO)

For the following modifications:

- Different manufacturer
- Different material, reduction in material purity
- Change in production process

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12)
- Damp heat test (MQT 13)

Repeat for IEC 61730:

- Impulse voltage test (MST 14)
- Reverse current overload test (MST 26)

The use of process gases or targets from different suppliers, but same minimum purities does not require retesting.

4.3.4 Modification to cell technology

For the following modifications:

- Change in cell structure (modification of any layer / additional layer / change of overall cell thickness > 10 %)
- Reduction in process gas purity (including mixing ratios)
- Change in source or target material or reduction in target purity
- Change in doping material composition

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- Damp heat test (MQT 13)

Repeat for IEC 61730:

- Temperature test (MST 21)
- Reverse current overload test (MST 26)

A change in the semiconductor material/technology requires a full qualification.

4.3.5 Modification to cell layout

For the following modifications:

- Change in number of cells (given that the output power increases by > 10 %)
- Change in serial/parallel connection
- Change in number of cross-scribes
- Change in cell width > 10 %
- Change in patterning technology (e.g. laser vs. mechanical)
- Increase in number of cells per bypass diode

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- Bypass diode thermal test (MQT 18) if the short-circuit current increases by > 10 %

Repeat for IEC 61730:

- Reverse current overload test (MST 26)

4.3.6 Modification to back contact

For the following modifications:

- Different manufacturer
- Different material, reduction in material purity
- Change in production process

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12)
- Damp heat test (MQT 13)

Repeat for IEC 61730:

- Impulse voltage test (MST 14)

- Reverse current overload test (MST 26)

The use of process gases or targets from different suppliers, but same minimum purities does not require retesting.

4.3.7 Modification to edge deletion

For the following modifications:

- Process (e.g. laser vs. mechanical process)
- Reduction in width (edge distance)

Repeat for IEC 61215:

- Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12)
- Damp heat test (MQT 13)

Repeat for IEC 61730:

- Impulse voltage test (MST 14)
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint
- Sequence B1 if design qualified for pollution degree 1

4.3.8 Modification to interconnect material or technique

For the following modifications:

- Different material (e.g. alloy, chemistry and core)
- Change in mechanical properties by more than 10 % of tensile strength, yield strength and elongation
- Change in thickness by more than 10 %
- Change in (total) cross-section of interconnect material by more than 10 %
- Different bonding technique
- Different bonding material (e.g. adhesive)
- Change in the number of interconnect or bonding points or decrease in bonding area per contact point
- Change in contact layout (e.g. position of cross connectors or internal conductors behind the cells)
- Change in insulation tape (thickness, material, manufacturer)

Repeat for IEC 61215:

- Thermal cycling test, 200 cycles (MQT 11)
- Damp heat test (MQT 13) for changes in material

Repeat for IEC 61730:

- Reverse current overload test (MST 26)

4.3.9 Modification to backsheet

A change from glass to non-glass or vice-versa requires a full qualification.

For the following modifications:

- Different material, i.e. any change in specification of the material or any of its layers

- Glass: reduction of thickness by more than 10 %; non-glass: change of thickness by more than 20 % of any one of the individual layers (while maintaining the required minimum distance through insulation)
- For glass, if there is a reduction in the strengthening process (e.g. retest if change is from tempered to heat strengthened or annealed glass)
- Different surface treatments (inside or outside)
- Change of amount of adhesives, primers or other additives
- Addition or removal of adhesives, primers or additives

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09) for glass if change in heat strengthening process or if thickness is reduced
- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) / Retention of junction box on mounting surface (MQT 14.1) (can be omitted for glass with identical UV cut-off; not for glass change). MQT 14.1 can be omitted if junction box is mounted on frontsheet.
- Damp heat test (MQT 13) (if non-glass or if cells are deposited on back glass; not for glass change) or if surface treatment is added/changed (inside or outside).
- Static mechanical load test (MQT 16) if glass (including change in manufacturer) or if mounting depends on adhesion to backsheet
- Hail test (MQT 17) if rigidity depends on backsheet

Repeat for IEC 61730:

- Insulation thickness test (MST 04) if non-glass
- Cut susceptibility test (MST 12) if non-glass
- Impulse voltage test (MST 14) if reduced thickness or if change in material
- Temperature test (MST 21) if non-glass and if change in material
- Ignitability test (MST 24) if non-glass
- Module breakage test (MST 32) if glass (can be omitted for different surface treatments that do not impair mechanical strength)
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint and if backsheet is part of it
- Materials creep test (MST 37) (not for reduction of thickness and not for different outside surface treatment)
- Sequence B if non-glass
- Sequence B1 if design qualified for pollution degree 1

For increased thickness, the Materials creep test (MST 37) is required.

Additionally, in case of colour change of backsheet potentially resulting in higher PV module operating temperatures, consider repetition of the Temperature test (MST 21), if applicable.

4.3.10 Modification to electrical termination

Components for electrical terminations such as junction box, cables and connectors shall meet the relevant IEC standards referenced in IEC 61730-1. Their combination with other components and materials shall be tested either on PV module level or on component level. A change in component combination may result in the following PV module tests in addition to tests listed in the relevant component standards.

For the following modifications:

- Different material
- Different design (e.g., different dimensions, changed positions, number of junction boxes)
- Different potting material
- Different method of mechanical attachment / securement (e.g. adhesive change)
- Different method of electrical attachment (e.g. solder, crimping, brazing, etc.)

Repeat for IEC 61215:

- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) / Robustness of terminations test (MQT 14.1 and 14.2) (UV preconditioning test can be omitted for change in potting material or in case junction box is not directly exposed to sunlight; Test of cord anchorage (MQT 14.2) can be omitted for change in mechanical attachment of junction box; Retention of junction box on mounting surface (MQT 14.1) can be omitted for change in electrical attachment of cables)
- Thermal cycling test, 200 cycles (MQT 11) only for change in electrical attachment
- Damp heat test (MQT 13)
- Bypass diode thermal test (MQT 18) (not required for change of any attachment)

Repeat for IEC 61730:

- Accessibility test (MST 11)
- Temperature test (MST 21) if change in potting material or adhesive
- Ignitability test (MST 24) only for change of adhesive
- Reverse current overload test (MST 26) (not for change of adhesive)
- Screw connections test (MST 33) if applicable
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint (only for mechanical attachment of junction box)
- Materials creep test (MST 37) only for change of adhesive or of junction box
- Sequence B only for change of adhesive
- Sequence B1 if design qualified for pollution degree 1

4.3.11 Modification to bypass diode

For bypass diodes mounted within the junction box, requirements of IEC 62790 need to be fulfilled.

For bypass diodes not mounted within the junction box, the following applies:

For the following modifications:

- Lower rating of diode current or diode junction temperature
- Different number of bypass diodes per PV module
- Different type of bypass diode
- Different manufacturer of bypass diode
- Different mounting method (physical configuration, soldering material, bonding process, soldering temperature or process)

Repeat for IEC 61215:

- Thermal cycling test, 200 cycles (MQT 11) only for different mounting method
- Bypass diode thermal test (MQT 18)

Repeat for IEC 61730:

- Reverse current overload test (MST 26) only for different mounting method

4.3.12 Modification to edge sealing

For the following modifications:

- Different material
- Different thickness or width

Repeat for IEC 61215:

- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) if edge sealing is outer enclosure
- Damp heat test (MQT 13)

Repeat for IEC 61730:

- Impulse voltage test (MST 14)
- Ignitability test (MST 24) (not for different thickness or width)
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint
- Sequence B (not for different thickness or width)
- Sequence B1 if design qualified for pollution degree 1

4.3.13 Modification to frame and/or mounting structure

For the following modifications:

- Shape and/or cross-section of frame
- Reduction of surface area in contact between laminate and frame per linear dimension
- Different material including adhesive or mounting material
- Different mounting method (as defined in installation manual)
- Change in frame corner design
- Change in frame adhesive
- Change from framed to frameless PV module or vice versa

Repeat for IEC 61215:

- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12) if mounting relies on adhesive or polymeric framing material
- Thermal cycling test, 200 cycles (MQT 11) if mounting relies on adhesive or polymeric framing material
- Damp heat test (MQT 13) if mounting relies on adhesive or polymeric framing material or if change from framed to frameless PV module or vice versa
- Static mechanical load test (MQT 16)
- Hail test (MQT 17) if polymeric frame / frontsheet or if change from framed to frameless PV module

Repeat for IEC 61730:

- Continuity test of equipotential bonding (MST 13) if change in method of assembly (can be omitted if change in adhesive)
- Ignitability test (MST 24) for polymeric frames

- Module breakage test (MST 32)
- Screw connections test (MST 33) if applicable
- Material creep test (MST 37) if creep is not prevented by frame or other support anymore
- Sequence B for polymeric frames

In case of change of frame manufacturer or mounting system manufacturer (same material specifications and design) no tests are required.

4.3.14 Change in PV module size

For increase by more than 20 % of length, width or area

Repeat for IEC 61215:

- Thermal cycling test, 200 cycles (MQT 11) if change in interconnection layout (e.g. longer cross-connectors)
- Damp heat test (MQT 13)
- Static mechanical load test (MQT 16)
- Hail test (MQT 17) if non-tempered glass or if non-glass

Repeat for IEC 61730:

- Reverse current overload test (MST 26)
- Module breakage test (MST 32)

4.3.15 Higher or lower output power (by 10 % or more) with the identical design and size

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- Thermal cycling test, 200 cycles (MQT 11) if short-circuit current is increased by more than 10 %
- Bypass diode thermal test (MQT 18) if short-circuit current is increased by more than 10 %
- For performance at STC (MQT 06.1) see 4.1.

Repeat for IEC 61730:

- Reverse current overload test (MST 26)

4.3.16 Increase of over-current protection rating

Repeat for IEC 61730:

- Continuity test of equipotential bonding (MST 13)
- Reverse current overload test (MST 26)

4.3.17 Increase of system voltage

Repeat for IEC 61215:

- Hot-spot endurance test (MQT 09)
- UV preconditioning test (MQT 10) / Thermal cycling test, 50 cycles (MQT 11) / Humidity freeze test (MQT 12)
- Thermal cycling test, 200 cycles (MQT 11)
- Damp heat test (MQT 13)

Repeat for IEC 61730:

- Insulation thickness test (MST 04)
- Accessibility test (MST 11)
- Cut susceptibility test (MST 12) if non-glass
- Continuity test of equipotential bonding (MST 13)
- Impulse voltage test (MST 14)
- Peel test (MST 35) or Lap shear strength test (MST 36) if design includes cemented joint
- Sequence B

Annex A (informative)

A.1 Required retests for crystalline silicon PV modules, tabular overview

Table A.1 – Required retests for crystalline silicon PV modules

	IEC test	IEC test No.
	Hot-spot endurance test	MQT 09
	UV preconditioning test	MQT 10
	Thermal cycling test (50)	MQT 11
	Thermal cycling test (200)	MQT 11
	Humidity-freeze test	MQT 12
	Damp heat test	MQT 13
	Retention of junction box on mounting surface	MQT 14.1
	Test of cord anchorage	MQT 14.2
	Static mechanical load test	MQT 16
	Hail test	MQT 17
	Bypass diode testing	MQT 18
	Insulation thickness test	MST 04
	Accessibility test	MST 11
	Cut susceptibility test	MST 12
	Continuity test of equipotential bonding	MST 13
	Impulse voltage test	MST 14
	Temperature test	MST 21
	Ignitability test	MST 24
	Reverse current overload test	MST 26
	Module breakage test	MST 32
	Screw connections test	MST 33
	Peel test	MST 35
	Lap shear strength test	MST 36
	Materials creep test	MST 37
	Sequence B	MST 52-54
	Sequence B1	MST 52/55/56
Sub-clause	Modification (short term)	
4.2.1	Frontsheet	
4.2.2	Encapsulation	
4.2.3	Cell	
4.2.4	Interconnection	
4.2.5	Backsheet	
4.2.6	El. termination	
4.2.7	Diode	

		IEC test	
IEC test No.			
4.2.8	Circuitry	MQT 09	Hot-spot endurance test
4.2.9	Edge sealing	MQT 10	UV preconditioning test
4.2.10	Frame	MQT 11	Thermal cycling test (50)
4.2.11	Module size	MQT 11	Thermal cycling test (200)
4.2.12	Output power	MQT 12	Humidity-freeze test
4.2.13	Reverse current	MQT 13	Damp heat test
4.2.14	System voltage	MQT 14.1	Retention of junction box on mounting surface
4.2.15	Cell fixing tape	MQT 14.2	Test of cord anchorage
Test necessary. If applicable. Check detailed specification in 4.2.		MQT 16	Static mechanical load test
		MQT 17	Hail test
		MQT 18	Bypass diode testing
		MST 04	Insulation thickness test
		MST 11	Accessibility test
		MST 12	Cut susceptibility test
		MST 13	Continuity test of equipotential bonding
		MST 14	Impulse voltage test
		MST 21	Temperature test
		MST 24	Ignitability test
		MST 26	Reverse current overload test
		MST 32	Module breakage test
		MST 33	Screw connections test
		MST 35	Peel test
		MST 36	Lap shear strength test
		MST 37	Materials creep test
		MST 52-54	Sequence B
		MST 52/55/56	Sequence B1

A.2 Required retests for thin-film PV modules, tabular overview

Table A.2 – Required retests for thin-film silicon PV modules

Sub-clause	IEC test No.	IEC test	
		Hot-spot endurance test	UV preconditioning test
4.3.1	MQT 09		
4.3.2	MQT 10		
4.3.3	MQT 11		
4.3.4	MQT 11		
4.3.5	MQT 12		
4.3.6	MQT 13		
4.3.7	MQT 14.1		
4.3.8	MQT 14.2		
4.3.9	MQT 16		
4.3.10	MQT 17		
4.3.11	MQT 18		
4.3.12	MST 04		
	MST 11		
	MST 12		
	MST 13		
	MST 14		
	MST 21		
	MST 24		
	MST 26		
	MST 32		
	MST 33		
	MST 35		
	MST 36		
	MST 37		
	MST 52-54		
	MST 52/55/56		

		IEC test	IEC test No.
4.3.13	Frame		MQT 09
			MQT 10
4.3.14	Module size		MQT 11
			MQT 11
4.3.15	Output power		MQT 12
			MQT 13
4.3.16	Reverse current		MQT 14.1
			MQT 14.2
4.3.17	System voltage		MQT 16
			MQT 17
			MQT 18
			MST 04
			MST 11
			MST 12
			MST 13
			MST 14
			MST 21
			MST 24
			MST 26
			MST 32
			MST 33
			MST 35
			MST 36
			MST 37
			MST 52-54
			MST 52/55/56
Test necessary.			
If applicable. Check detailed specification in 4.3.			

A.3 Combined test flow IEC 61215 and IEC 61730 (see Figure A.1 and Table A.3)

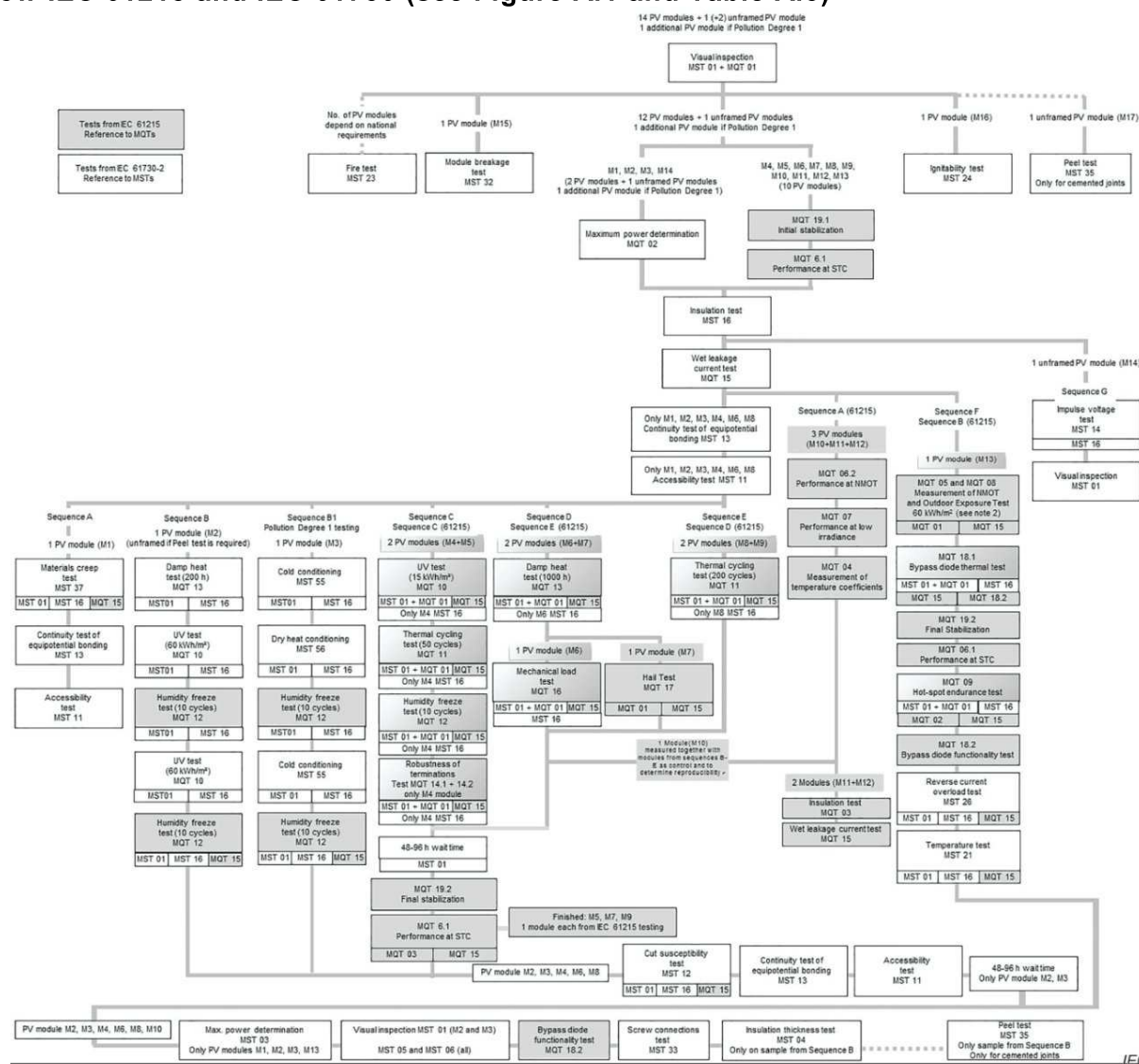


Figure A.1 – Combined test flow IEC 61215 and IEC 61730

Table A.3 – IEC identifiers for test sequences

IEC identifier	Type of sample	Standard covered	Test sequence
M1	PV module	IEC 61730	IEC 61730 Sequence A
M2	Unframed PV module	IEC 61730	IEC 61730 Sequence B
M3	PV module	IEC 61730	IEC 61730 Sequence B1
M4	PV module	IEC 61215 + IEC 61730	IEC 61215 Sequence C + IEC 61730 Sequence C
M5	PV module	IEC 61215 + IEC 61730	IEC 61215 Sequence C + IEC 61730 Sequence C
M6	PV module	IEC 61215 + IEC 61730	IEC 61215 Sequence E + IEC 61730 Sequence D
M7	PV module	IEC 61215 + IEC 61730	IEC 61215 Sequence E + IEC 61730 Sequence D
M8	PV module	IEC 61215 + IEC 61730	IEC 61215 Sequence D + IEC 61730 Sequence E
M9	PV module	IEC 61215 + IEC 61730	IEC 61215 Sequence D + IEC 61730 Sequence E
M10	PV module	IEC 61215	IEC 61215 Sequence A + IEC 61730 Control
M11	PV module	IEC 61215	IEC 61215 Sequence A
M12	PV module	IEC 61215	IEC 61215 Sequence A
M13	PV module	IEC 61215 + IEC 61730	IEC 61215 Sequence B + IEC 61730 Sequence F
M14	Unframed PV module	IEC 61730	IEC 61730 Sequence G
M15	PV module	IEC 61730	Module breakage test
M16	PV module	IEC 61730	Ignitability test
M17	Unframed PV module	IEC 61730	Peel test reference sample, if applicable
Tbd	PV module	IEC 61730	Fire test samples

A.4 Tests for new combinations of material in direct contact with each other

If more than one component is newly qualified and shall be used as an alternative to other already qualified components, care has to be taken that the performed tests cover all specific material combination.

EXAMPLE If a PV module design with e.g. encapsulant (A) and cell (A) is qualified and new alternative materials such as e.g. encapsulant (B) and cell (B) are introduced, all possible combinations shall be evaluated if each material can individually substitute another one. Each combination shall be tested following the requirements of this document. This results in the following 3 steps (see also Table A.4 and Figure A.2 below for test details):

- a) For qualification of new encapsulant (B) and new cell (B), see below combination A: use the more compulsive test regulation (“test necessary” supersedes “if applicable”, etc.) of encapsulation or cell.
- b) For additional qualification of new encapsulant (B) with cell (A), see below combination B: all encapsulation tests are required.
- c) For additional qualification of new cell (B) with encapsulant (A), see below combination C: all cell tests are required.

Table A.4 – Example for required tests for new material combinations

		IEC test	
IEC test No.			
Sub-clause Modification (short term)			
New combination A:			
New combination B:			
New combination C:			
4.2.2 (B), new	Encapsulation (B), new		
	Cell (A)		
4.2.3	Required tests		
4.2.2 (A)	Encapsulation (A)		
	Cell (B), new		
4.2.3	Required tests		
	Test necessary.		
If applicable. Check detailed specification in 4.2.			

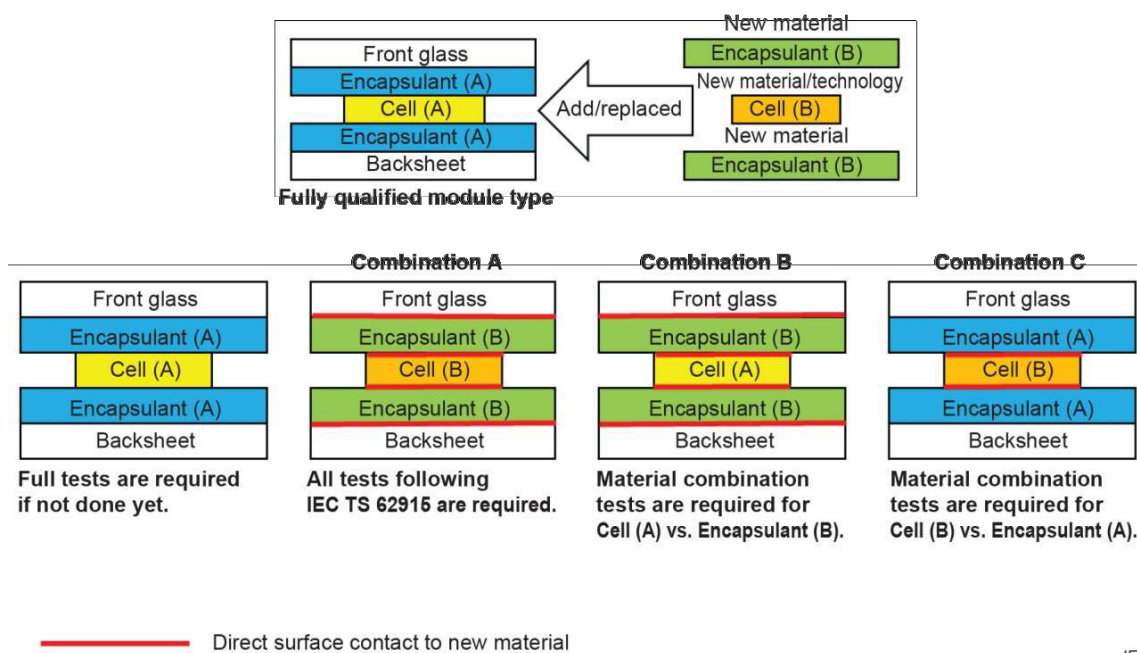


Figure A.2 – Illustration of example for required tests for new material combinations

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch